

7. Inos is a Welsh caravan manufacturer based in Denbighshire. The materials used to build modern caravans are very different to those used in early caravans.

**Table 1** shows how the materials used in manufacturing the body of a caravan have changed over time.

| Year | Materials used                            |
|------|-------------------------------------------|
| 1920 | wood                                      |
| 1950 | steel and wood                            |
| 1970 | aluminium and wood                        |
| 2015 | glass reinforced plastic (GRP), polyester |

**Table 2** shows information about some of the materials used to make the body of caravans.

| Material  |              | Density (g/cm <sup>3</sup> ) | Stiffness (GPa) | Melting Point (°C) | Strength (MPa) | Is it brittle? | Does it corrode?                  |
|-----------|--------------|------------------------------|-----------------|--------------------|----------------|----------------|-----------------------------------|
| wood      |              | 0.6                          |                 |                    | 1 000          | No             | no but rots when wet              |
| steel     |              | 7.8                          | 210             | 1 357              | 1 200          | No             | yes producing rust                |
| aluminium |              | 2.7                          | 69              | 660                | 90             | No             | yes producing a resistant coating |
| GRP       | glass fibres | 2.4                          | 180             | 1 400              | 3 500          | Yes            | No                                |
|           | epoxy resin  | 1.5                          |                 |                    | 60             | Yes            | No                                |
| polyester |              | 1.9                          | 150             | 121                | 250            | Yes            | No                                |

Use the information above to answer the following questions.

- (a) Describe how the weight and strength of caravan bodies changed between 1920 and 1970. [4]

Weight: .....

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Strength: .....

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